
ITCS3166

Computer Networks

Lecture 4 Medium Access Control

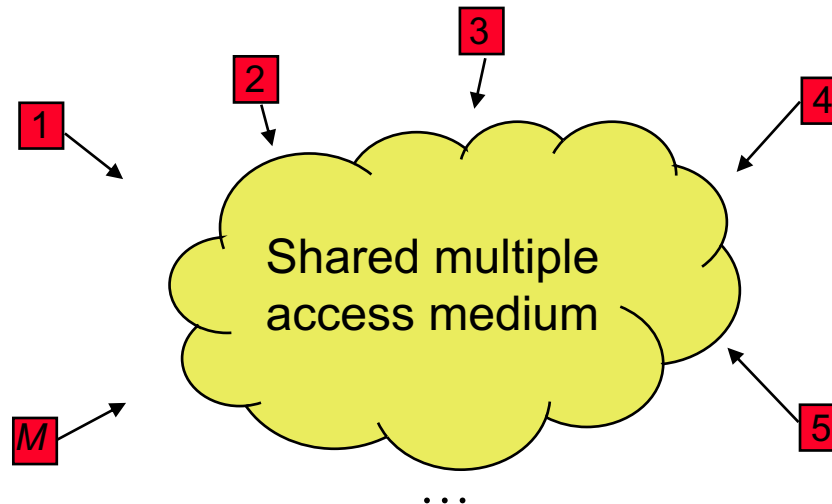
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Chapter Overview

- **Broadcast Networks**
 - **All information sent to all users**
 - **No routing**
 - **Shared media**
 - **Radio**
 - **Cellular telephony**
 - **Wireless LANs**
 - **Copper & Optical**
 - **Ethernet LANs**
 - **Cable Modem Access**
- *Medium Access Control*
 - **To coordinate access to shared medium**
 - **Data link layer since direct transfer of frames**
- *Local Area Networks*
 - **High-speed, low-cost communications between co-located computers**
 - **Typically based on broadcast networks**
 - **Simple & cheap**
 - **Limited number of users**

Multiple Access Communications

- **Shared media basis for broadcast networks**
 - Inexpensive: radio over air; copper or coaxial cable
 - M users communicate by broadcasting into medium
- **Key issue: How to share the medium (multi-access) when there is a competition for it?**
 - The control protocols are in MAC sublayer



Approaches to Media Sharing

Medium sharing techniques

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graph TD; A[Medium sharing techniques] --> B[Static channelization]; A --> C[Dynamic medium access control]; C --> D[Scheduling]; C --> E[Random access];
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Static channelization

- **Partition medium**
- **Dedicated allocation to users**
- **Satellite transmission**
- **Cellular Telephone**

Dynamic medium access control

Scheduling

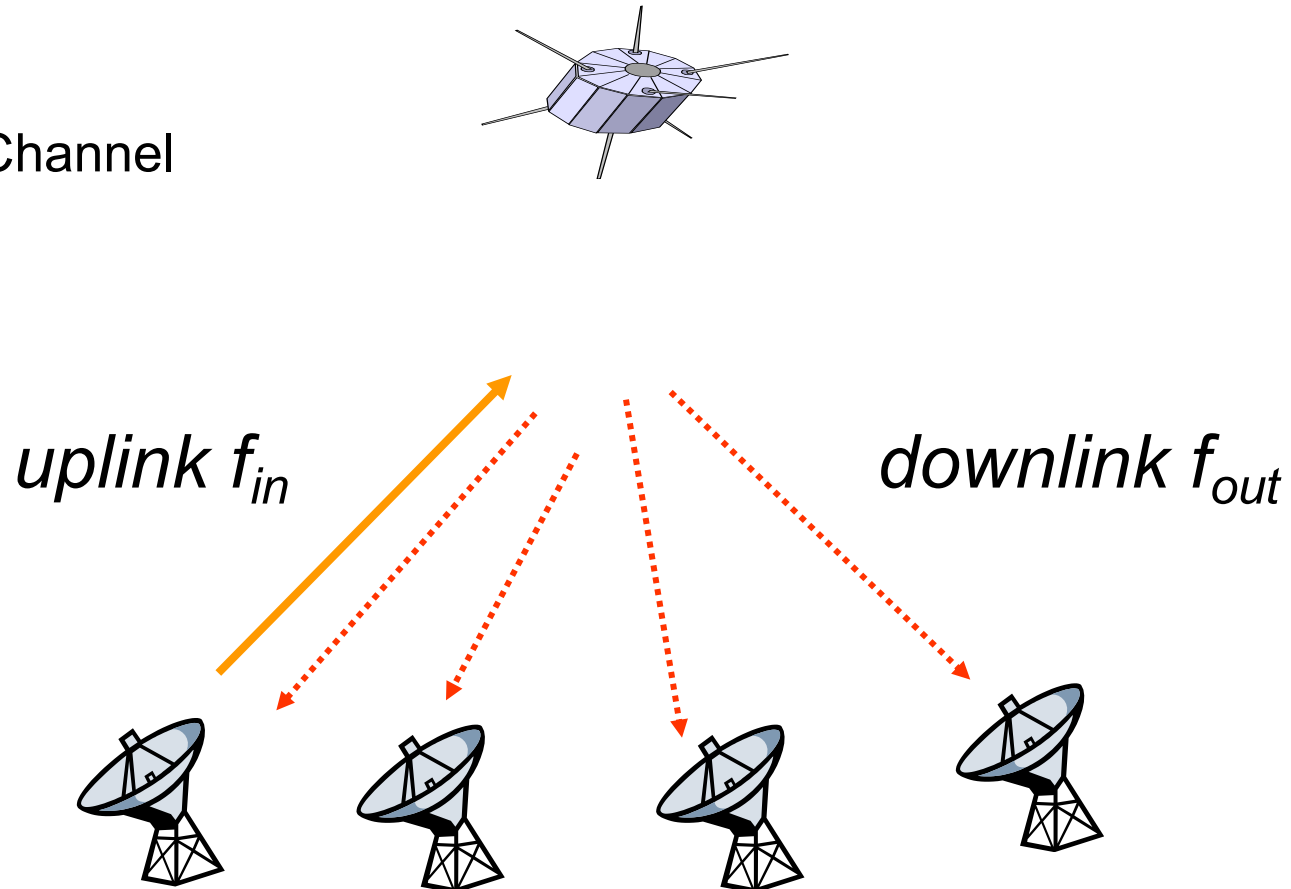
- **Polling: take turns**
- **Request for slot in transmission schedule**
- **Token ring**
- **Wireless LANs**

Random access

- **Loose coordination**
- **Send, wait, retry if necessary**
- **Aloha**
- **Ethernet**

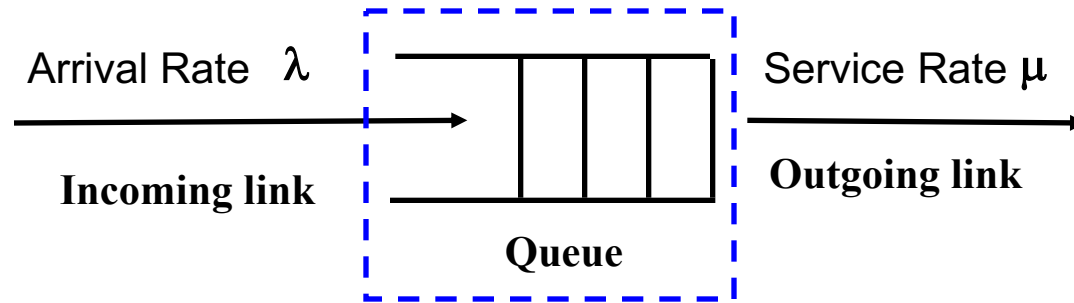
Channelization: Satellite

Satellite Channel



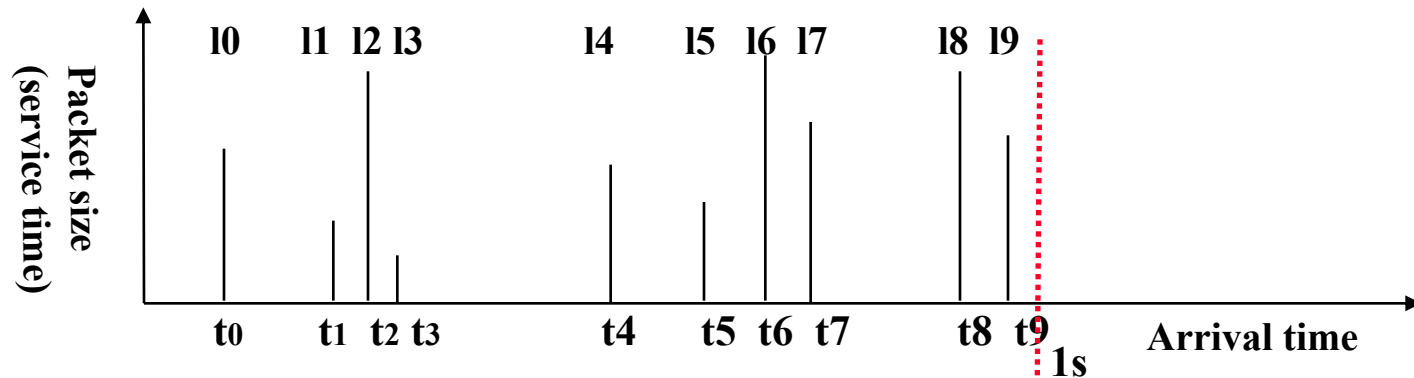
What we can learn from the foundations of queueing theory?

A General Queueing Model in Packet Network



- **Arrival rate λ :**
 - The number of packets coming from the incoming link in a unit time
- **Service rate μ :**
 - The number of packets forwarded by the outgoing link in a unit time
- **Utilization $\rho = \lambda / \mu$**
 - If $\rho > 1$, the queue's length is to be infinite, so is the queueing delay
 - If $\rho = 1$, the queue's length is to be infinite, so is the queueing delay if the inter-arrival distributions are not deterministic; otherwise, finite queue's length, **D/D/1** queue
 - **If $\rho < 1$, finite queue's length, finite queueing delay!**
 - **Most scenarios belong to this case !**

Preliminary of Queueing Theory



- **Three Components**
 - Inter-arrival time distributions ($t_0, t_1-t_0, t_2-t_1, t_3-t_2, \dots, t_9-t_8$)
 - Packet length (service time) distributions (10, 11, 12, 13, ..., 19)
 - The number of servers (**FIFO** queue discipline vs. PS)
- **Distributions**
 - Deterministic distribution (D)
 - Exponential distribution (M): (ce^{-cx}) – Poisson distribution
 - General distribution (like pareto...)
- **Response time (total time) = queueing delay + service time**

Preliminary of Queueing Theory (cont.)

- M/D/1 queue: arrival rate λ , service rate μ ($1/b$, where b is the fixed uniform service time)

$$\text{Average queueing-delay: } E(T_w) = \frac{1}{2\lambda} \cdot \frac{\rho^2}{(1-\rho)}$$

$$\text{Average response time } (T = T_w + T_s) = \frac{1}{2\lambda} \cdot \frac{\rho^2}{(1-\rho)} + b$$

- M/M/1 queue: arrival rate λ , service rate μ

$$\text{Average queueing-delay: } E(T_w) = \frac{\rho / \mu}{1 - \rho} = \frac{\rho}{\mu - \lambda}$$

$$\text{Average response time } (T = T_w + T_s) = \frac{1}{\mu(1 - \rho)} = \frac{1}{\mu - \lambda}$$

- Example: in a M/M/1 queue, given arrival rate $\lambda = 950$ / s, $\mu = 1000$ /s, what is the average queueing delay of packets?

$$\text{Utilization } \rho = \lambda / \mu$$

Static Channel Allocation in LANs and MANs

- Frequency Division Multiplexing (FDM): for N users, divide the bandwidth of the channel into N equal-sized portions
- Given a multi-access channel with capacity C bps, the average arrival rate of N users λ frames/sec (Poisson), average frame size $1/\mu$ bits/frame (exponential).

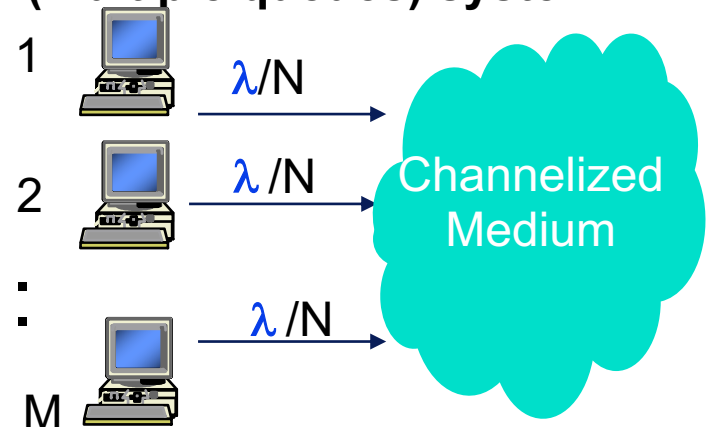
What is the service rate?

What is the average response time ($= 1 / (\text{service rate} - \text{arrival rate})$) of frames in a single channel (queue) system?

What is the average response time ($= 1 / (\text{service rate} - \text{arrival rate})$) of frames in a multi-channel (multiple queues) system?

When they are the same?

What is the problem of FDM?



Dynamic Channel Allocation in LANs and MANs

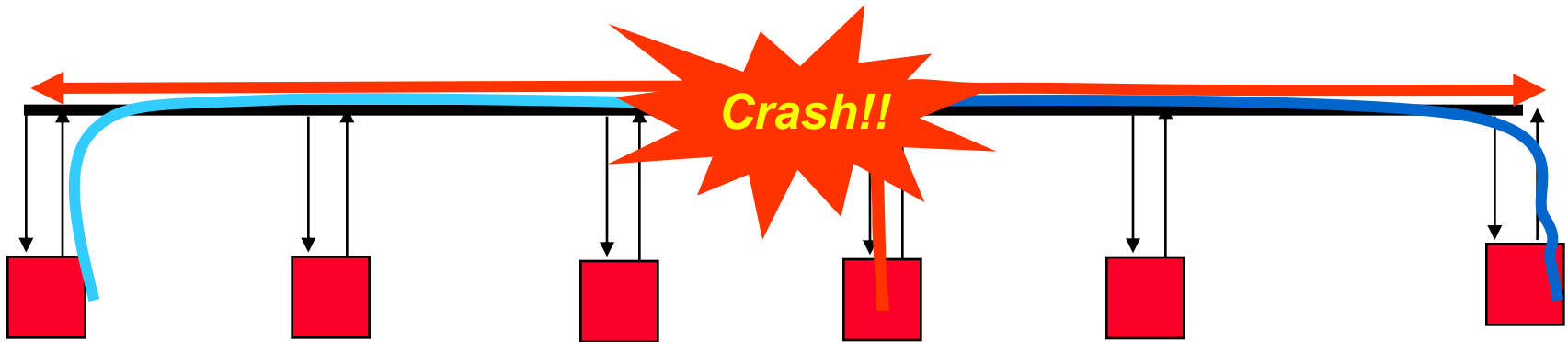
Assumptions

- **Station Model:** N independent stations generating frames at a constant rate, one-by-one transmission
- **Single Channel:** only one for all stations equivalently
- **Collision:** if two frames are transmitted simultaneously, the resulting signal is garbled
- (a) **Continuous Time:** transmission begins at any instant.
(b) **Slotted Time:** transmission begins at the start of a slot.
- (a) **Carrier Sense:** stations can sense the channel busy or idle.
(b) **No Carrier Sense:** just go ahead transmission

LANs generally have carrier sense, how about wireless networks?

Random Access

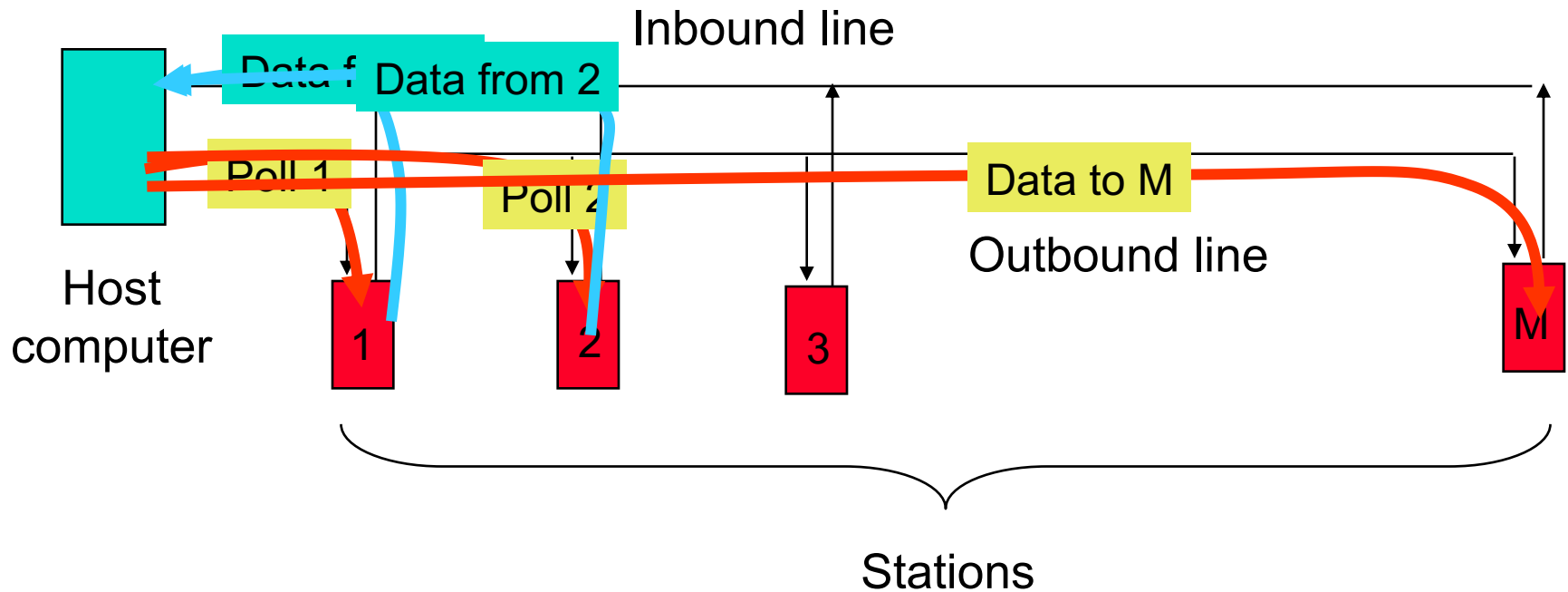
Multi-tapped (multit-access) Bus



Transmit when ready

Transmissions can occur; need retransmission strategy

Scheduling: Polling

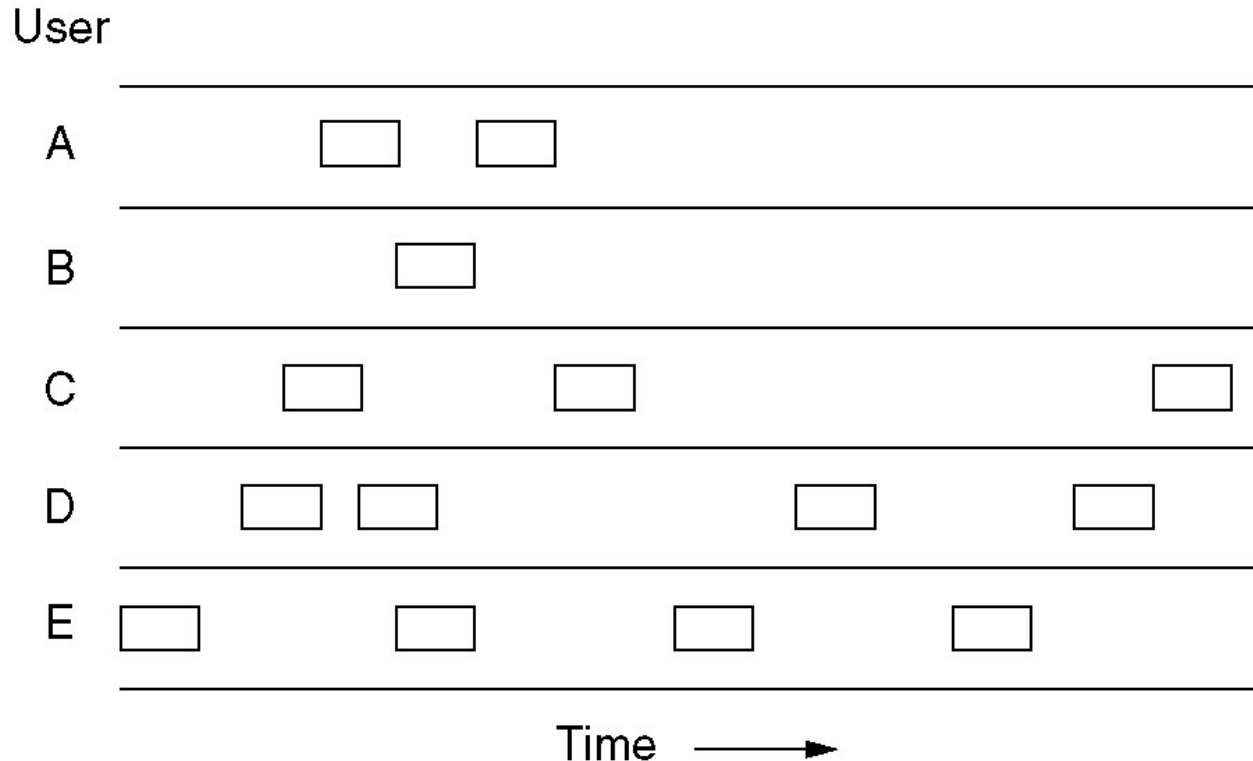


Multiple Access Protocols

- **ALOHA**
- **Carrier Sense Multiple Access Protocols**
- **Collision-Free Protocols**
- **Limited-Contention Protocols**
- **Wavelength Division Multiple Access Protocols**
- **Wireless LAN Protocols**

Pure ALOHA

In pure ALOHA, frames are transmitted at completely arbitrary times.



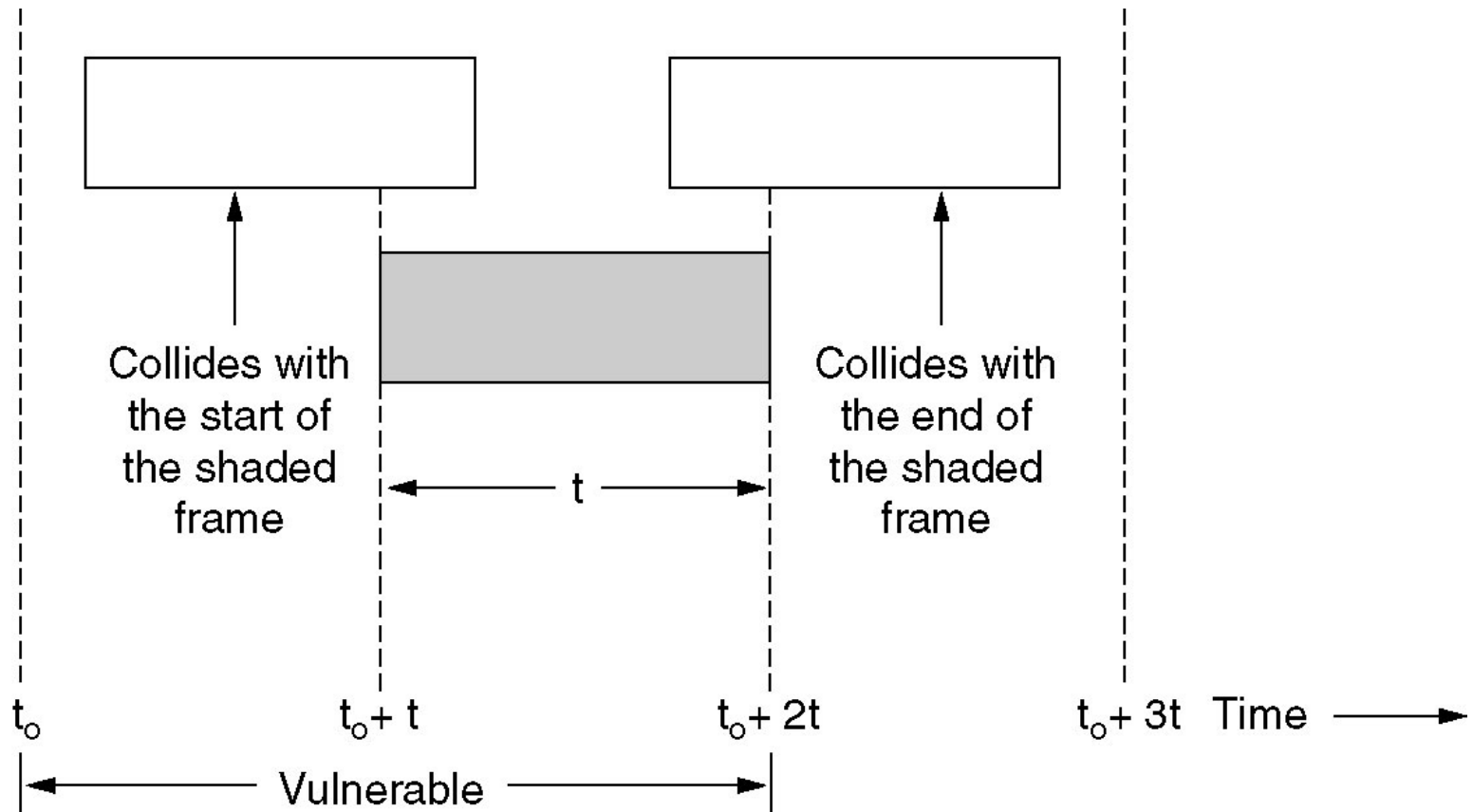
How a station knows its frame was destroyed due to collision?

Feedback property of broadcasting.

What is the efficiency of an ALOHA channel?

Pure ALOHA (2)

- G per frame time be the mean number of transmission attempts per frame time, and P_0 is the probability of a transmission does not suffer a collision, throughput $S = GP_0$



Vulnerable period for the shaded frame.

Slotted ALOHA

- **A station is required to wait for the beginning of the next slot, one time slot corresponding to one frame time**

What is the vulnerable period for a frame?

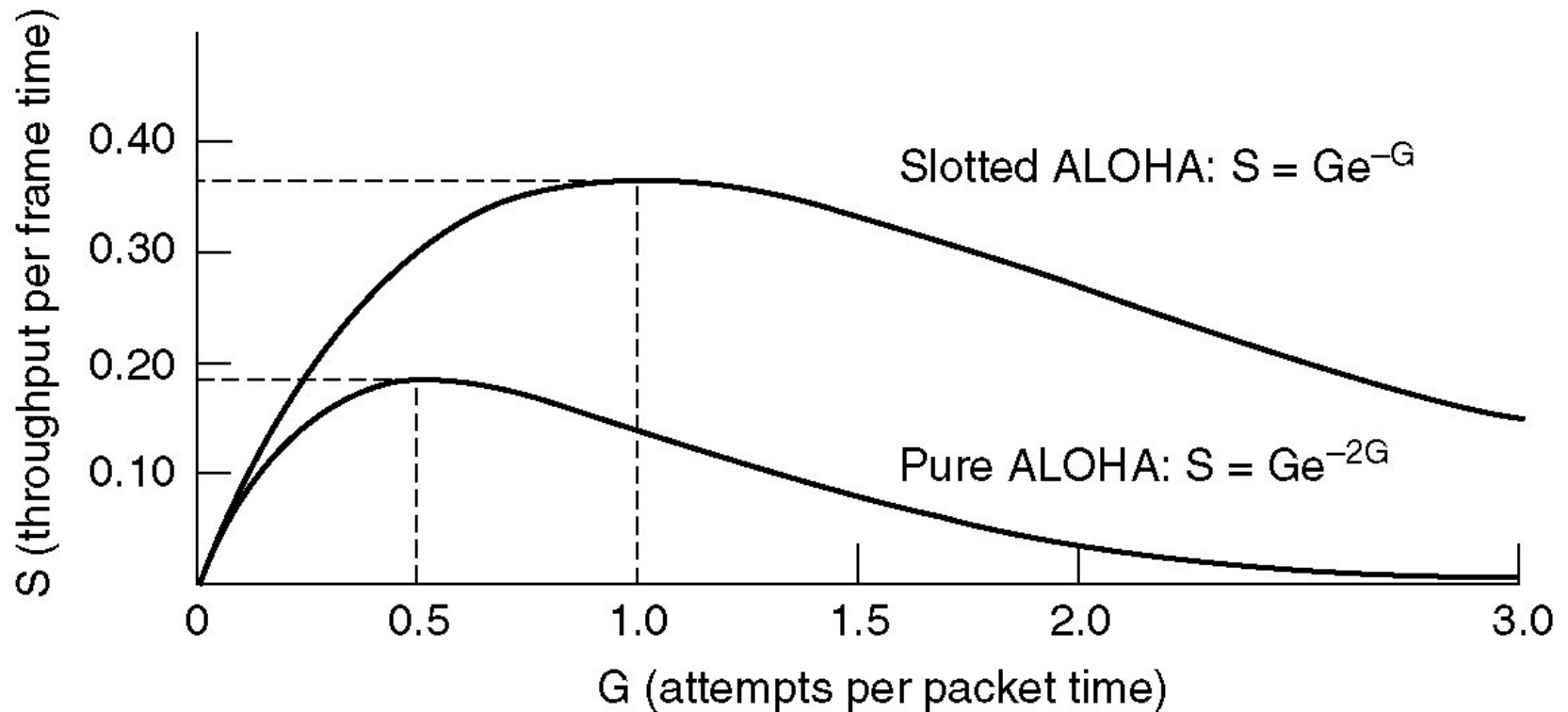
What is required to turn a pure ALOHA to a slotted ALOHA?

What benefit received by slotted ALOHA?

What is the price for the benefit?

Throughput of ALOHA Systems

Throughput versus offered traffic for ALOHA systems.



Are ALOHA protocols carrier sense?

Question?

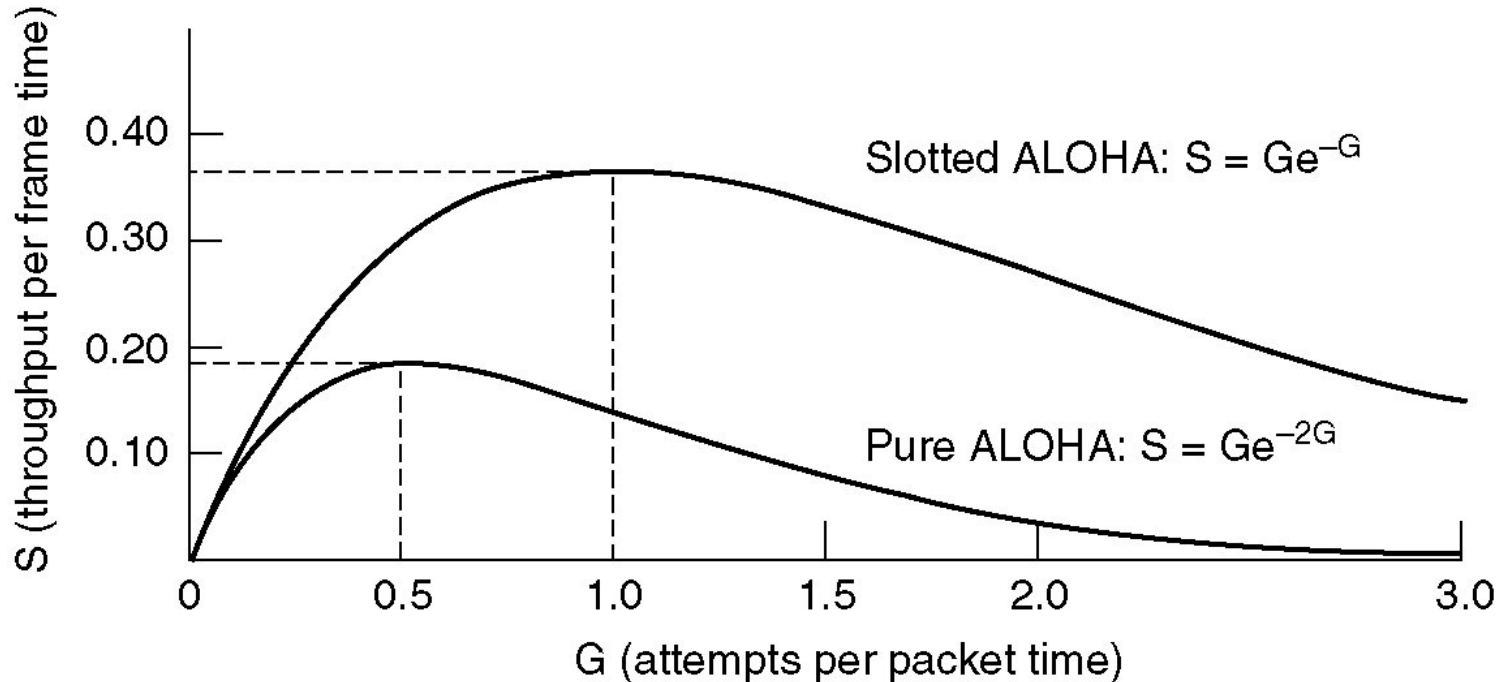
- ° **Consider the delay of pure ALOHA versus slotted ALOHA at low load. Which one is less? Explain your answer.**

With pure ALOHA transmission can start immediately. With slotted ALOHA, transmission has to wait to the next slot.

Example

- Five thousand banking stations are competing for the use of a single slotted ALOHA channel. The average station makes 72 requests/hour. A slot is 125 μ sec. What is the average total channel load? And what is the throughput **S**?

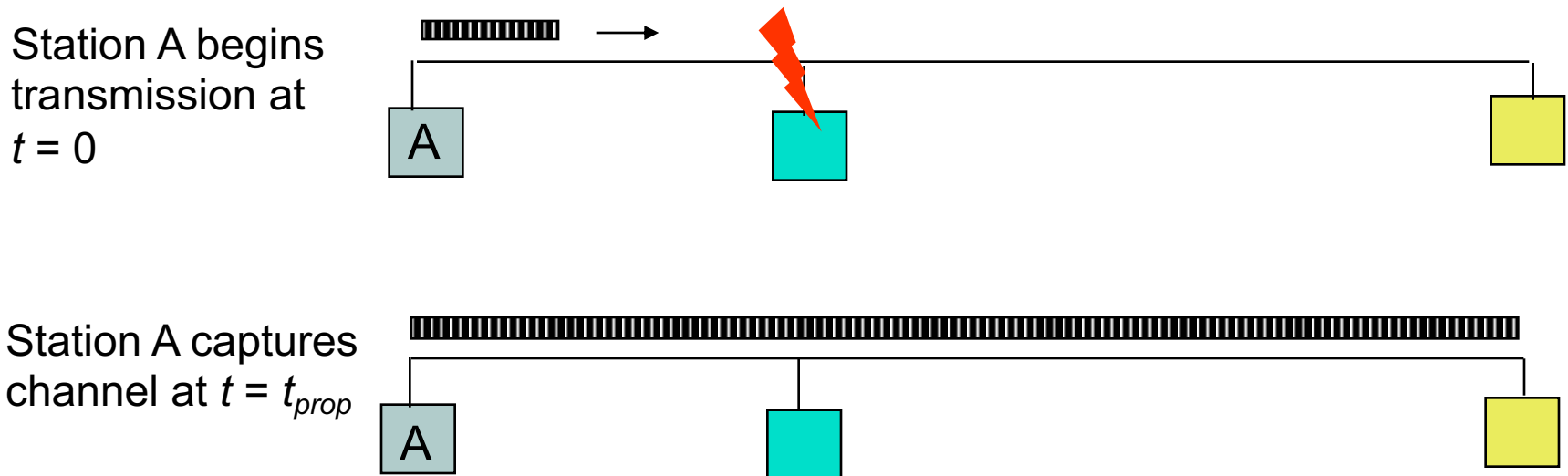
$$G = 5000 \times 0.02 \times 0.00125 = 1/80$$



Comparison of the channel utilization versus load.

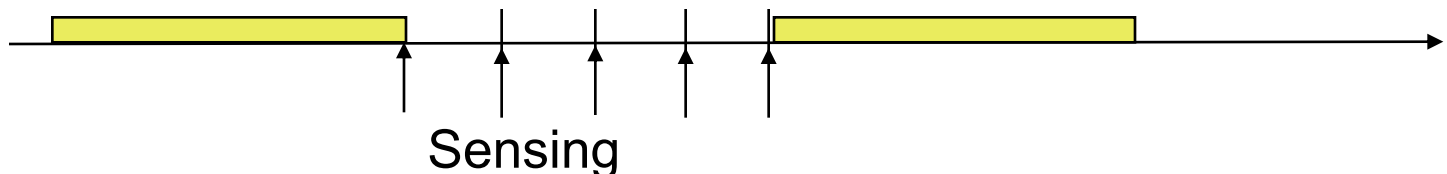
Carrier Sensing Multiple Access (CSMA)

- A station senses the channel before it starts transmission
 - If busy, either wait or schedule backoff (different options)
 - If idle, start transmission
 - **Vulnerable period is reduced to t_{prop} (channel capture effect)**
 - When collisions occur they involve entire frame transmission times
 - Always better than ALOHA?

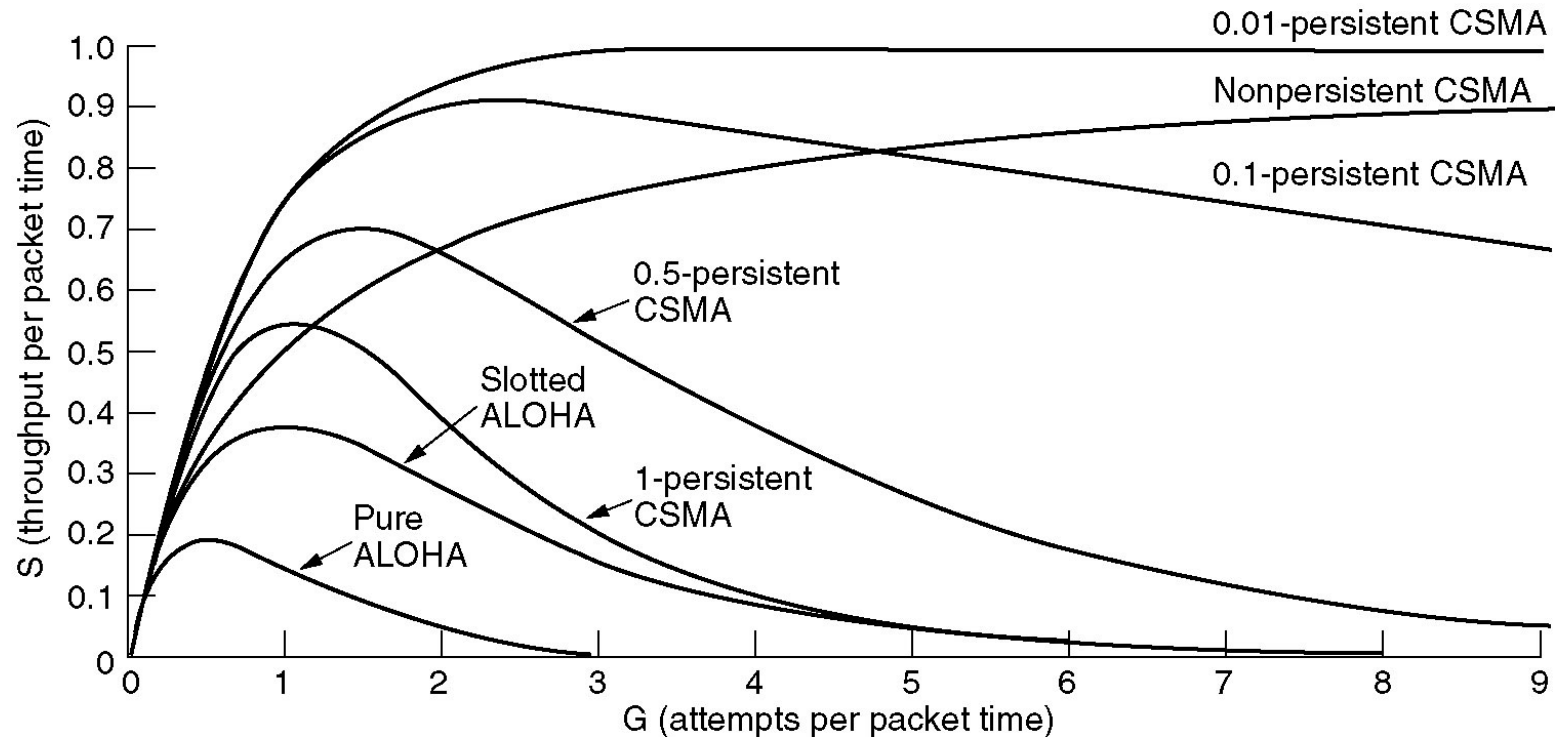


CSMA Options

- **Transmitter behavior when busy channel is sensed**
 - **1-persistent CSMA (most greedy)**
 - Start transmission as soon as the channel becomes idle
 - Possible simultaneously transmission in a propagation delay
 - **Low delay and low efficiency**
 - **Non-persistent CSMA (least greedy)**
 - If busy, wait a backoff period, then sense carrier again
 - **High delay and high efficiency**
 - **p-persistent CSMA (adjustable greedy)**
 - Wait till channel becomes idle, transmit with prob. p ; or wait one mini-slot time & re-sense with probability $1-p$
 - **Delay and efficiency can be balanced**



CSMA Performance



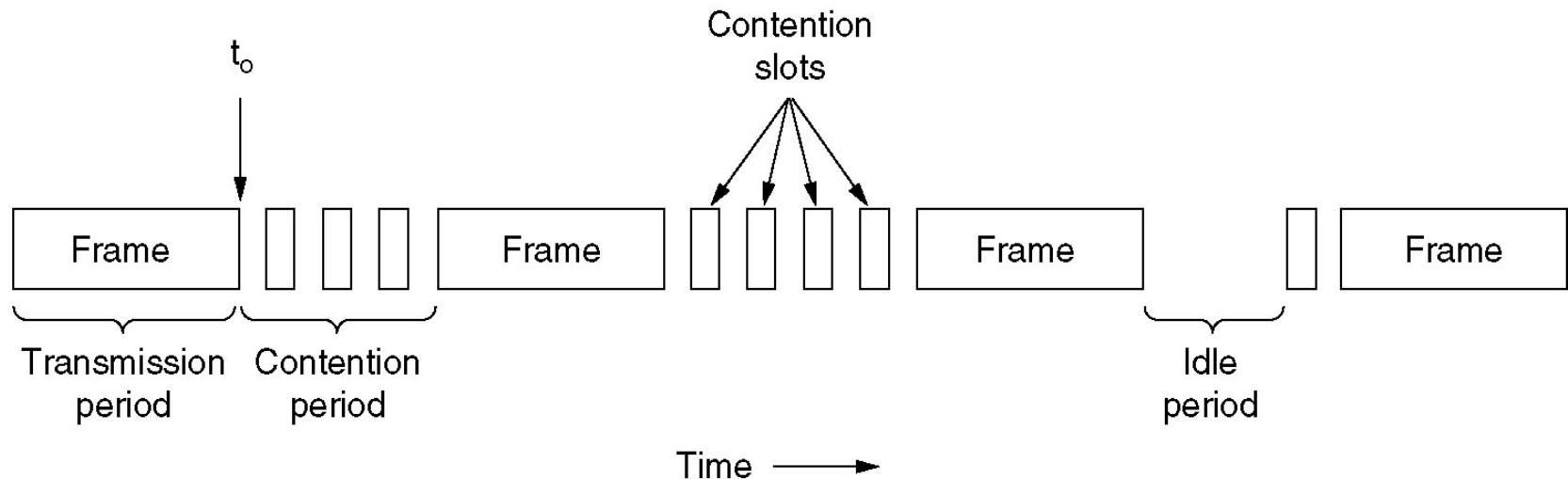
Comparison of the channel utilization versus load.

What is the price of non-persistent and p-persistent protocols?

Longer delay

CSMA with Collision Detection

- **CSMA/CD: a station aborts its transmission ASAP a detection of collision – quickly terminating a damaged frame saves **T& BW****
 - Widely used on LANs in the MAC sublayer



CSMA/CD can be in one of three states: contention, transmission, or idle.

Can collision occurs with CSMA/CD? If so, when occurs and not occurs?

When a station can seize the channel for transmission (how long is the contention)?

Why CSMA/CD More Efficient?

- **Monitor for collisions & abort transmission**
 - **Stations with frames to send, first do carrier sensing**
 - **After beginning transmissions, stations continue listening to the medium to detect collisions**
 - **If collisions detected, all stations involved stop transmission, reschedule random backoff times, and try again at scheduled times**
- **In CSMA collisions result in wastage of X seconds spent transmitting an entire frame**
- **CSMA-CD reduces wastage to time to detect collision and abort transmission**

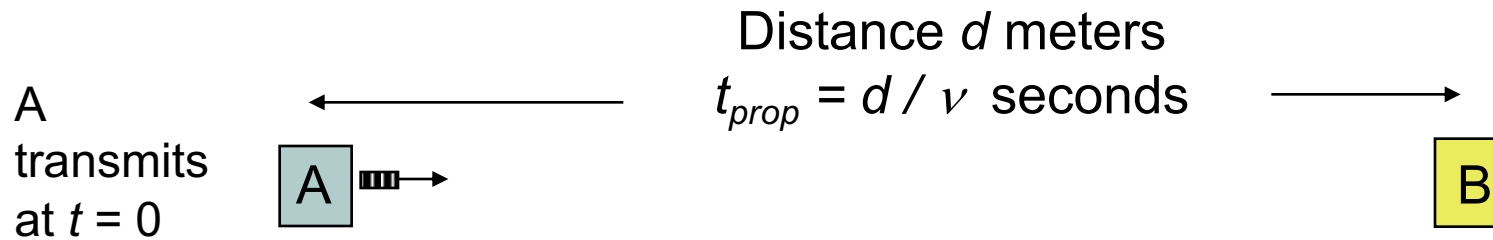
Delay-Bandwidth Product

- ***Delay-bandwidth* product key parameter**
 - Coordination in sharing medium involves using bandwidth (explicitly or implicitly)
 - Difficulty of coordination commensurate with delay-bandwidth product
- **Simple two-station example**
 - Station with frame to send listens to medium and transmits if medium found idle
 - Station monitors medium to detect collision
 - If collision occurs, station that begin transmitting earlier retransmits (propagation time is known)

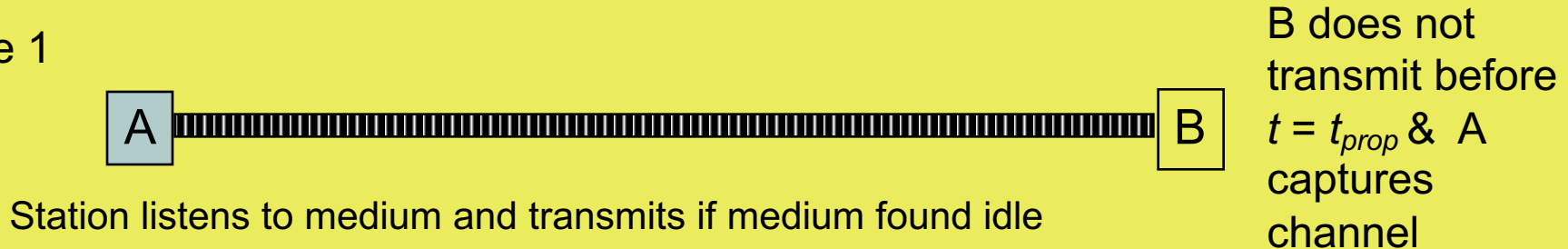
When a station can know it has seized the channel for transmission successfully?

Two-Station MAC Example

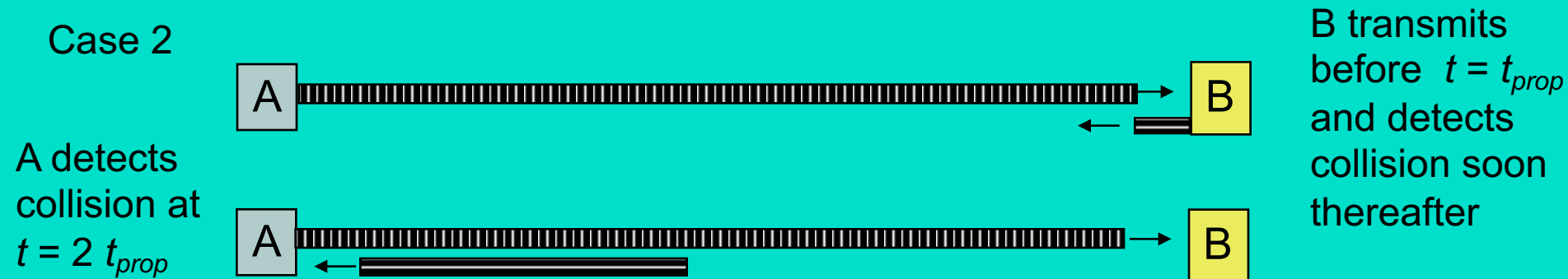
Two stations are trying to share a common medium



Case 1



Case 2



Efficiency of Two-Station Example

- Each frame transmission requires $2t_{prop}$ of quiet time
 - Station B needs to be quiet t_{prop} before *and* after time when Station A transmits
 - R transmission bit rate
 - L bits/frame

$$Efficiency = \rho_{\max} = \frac{L}{L + 2t_{prop}R} = \frac{1}{1 + 2t_{prop}R/L} = \frac{1}{1 + 2a}$$

$$MaxThroughput = R_{eff} = \frac{L}{L/R + 2t_{prop}} = \frac{1}{1 + 2a} R \text{ bits/second}$$

Normalized
Delay-Bandwidth
Product

$$a = \frac{t_{prop}}{L/R}$$

← Propagation delay

← Time to transmit a frame

Next: Scheduling for Medium Access Control

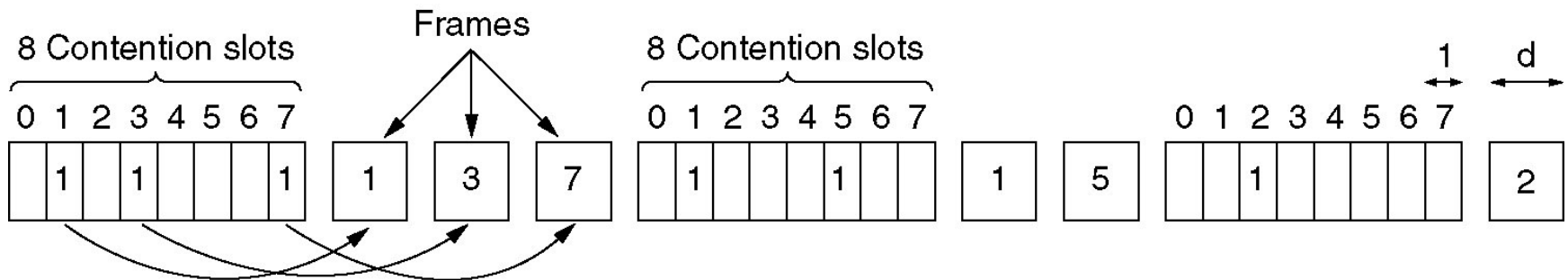
- **Schedule frame transmissions to avoid collision in shared medium**
 - ✓ **More efficient channel utilization**
 - ✓ **Less variability in delays**
 - ✓ **Can provide fairness to stations**
 - × **Increased computational or procedural complexity**
- **Two main approaches**
 - **Reservation**
 - **Polling**

Reservation System Options

- **Centralized or distributed system**
 - ***Centralized systems:*** A central controller listens to reservation information, decides order of transmission, issues grants
 - ***Distributed systems:*** Each station determines its slot for transmission from the reservation information
- **Single or Multiple Frames**
 - ***Single frame reservation:*** Only one frame transmission can be reserved within a reservation cycle
 - ***Multiple frame reservation:*** More than one frame transmission can be reserved within a frame
- **Channelized or Random Access Reservations**
 - ***Channelized (typically TDMA) reservation:*** Reservation messages from different stations are multiplexed without any risk of collision
 - ***Random access reservation:*** Each station transmits its reservation message randomly until the message goes through

Collision-Free – A Bit-Map Reservation Protocol

- Assumptions: **N** stations have unique addresses 0 to **N** -1
 - Which station gets the channel after a successful transmission?
- A bit-map protocol:
 - a contention period has exactly **N** slots and a station **j** announces it has a frame to send by inserting a bit of 1 into slot **j**



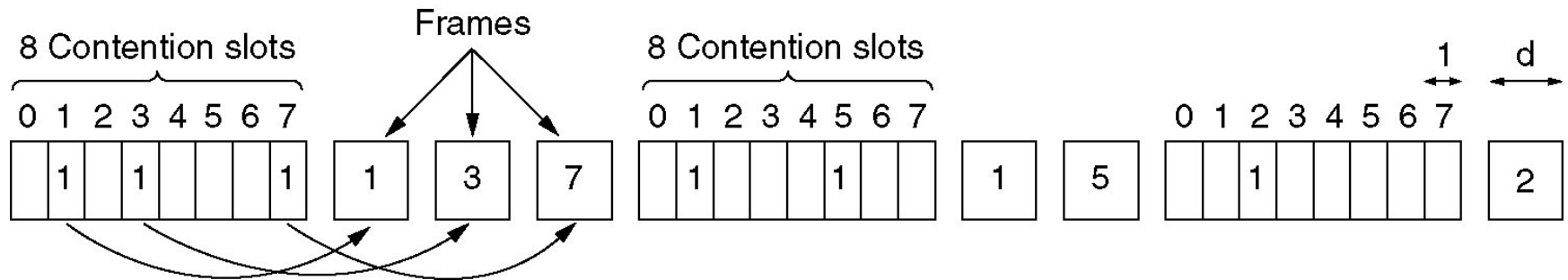
The basic bit-map protocol.

How long should be one contention slot?

Is it fair to stations with different addresses?

Analysis of a Bit-Map Protocol

- How long does a station have to wait in the worst case before it can start transmitting its frame over a LAN that uses the basic bit-map protocol?



The basic bit-map protocol.

What is the overhead per frame and the efficiency at high and low load?
(if propagation delay is negligible) how about delay?

Low load: $d/(N+d)$

High load: $d/(d+1)$

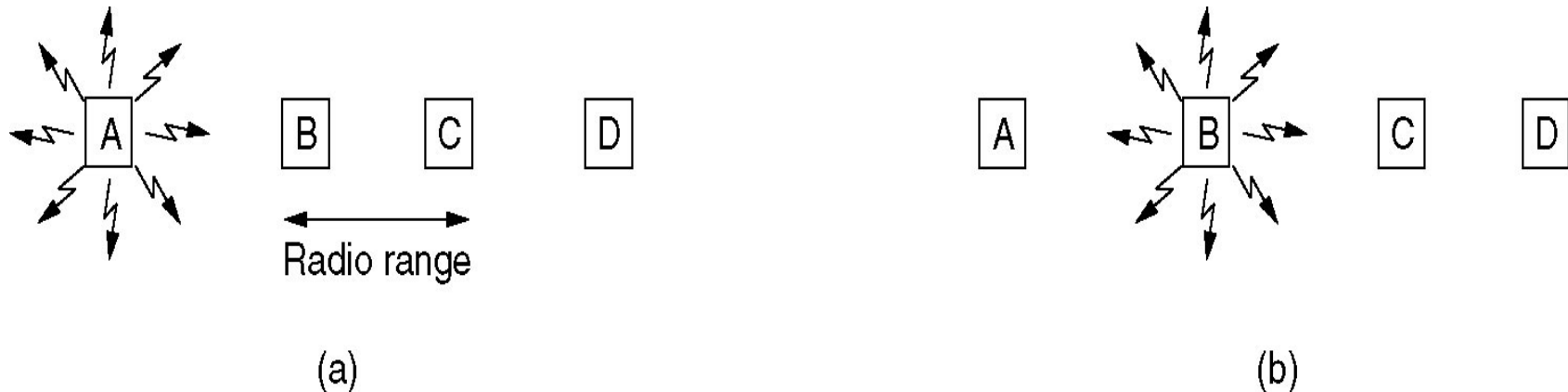
Question

- **How long does a station, s , have to wait in the worst case before it can start transmitting its frame over a LAN that uses the basic bit-map protocol?**

The worst case is where all stations want to send and s is the lowest-numbered station. Wait time N bit contention period + $(N - 1) \times d$ bit for transmission of frames. The total is $N + (N - 1)d$ bit times.

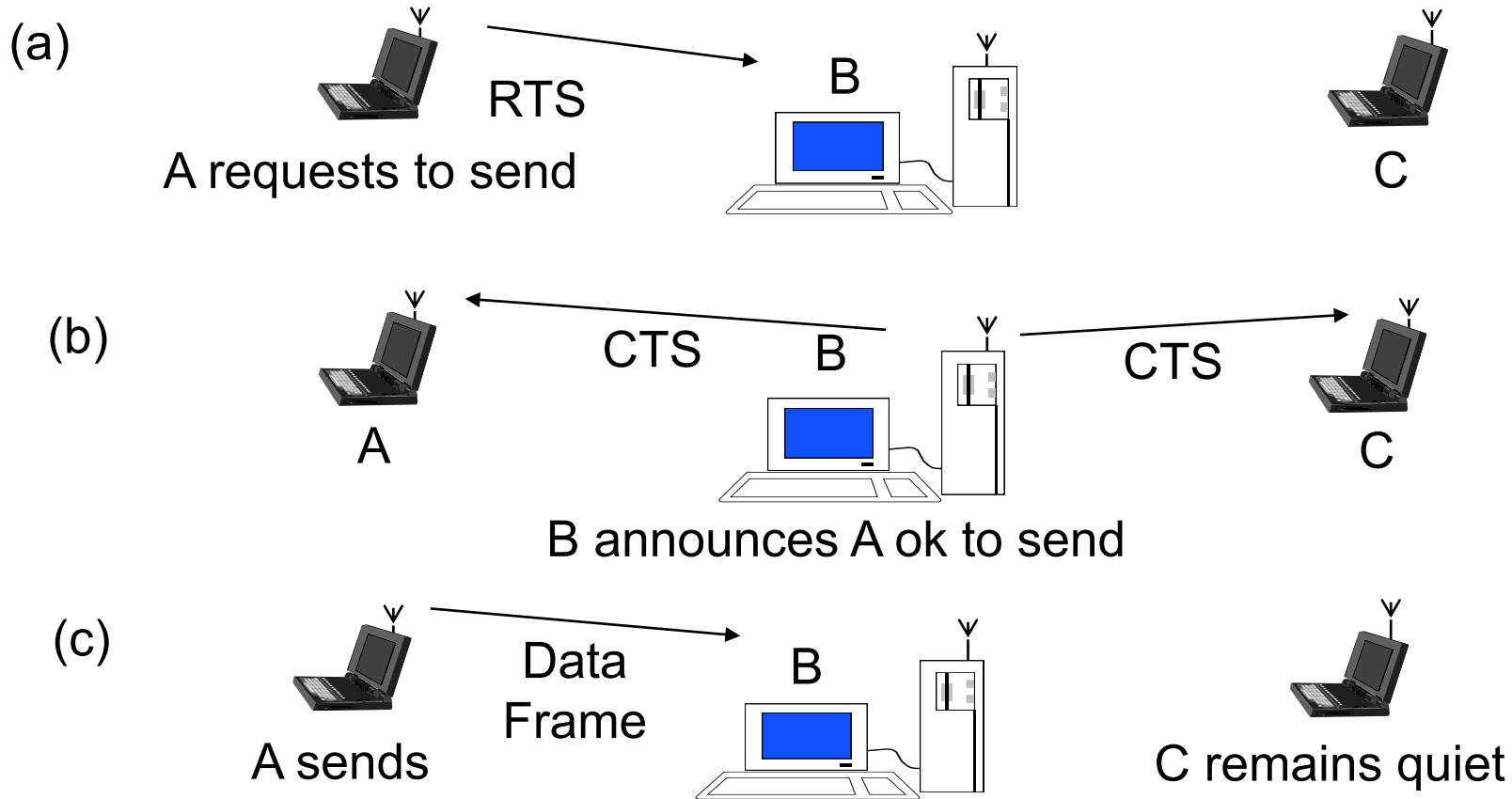
Wireless LAN Protocols

- **Assumptions: all radio transmitters have some fixed ranges**
 - If a receiver is within the range of two active transmitter, the resulting signal will be garbled and useless
- **Why not use CSMA (carrier-sense) in a wireless LAN?**
 - Hidden station problem and exposed station problem



A wireless LAN. (a) A transmitting to B. (b) B transmitting to A.

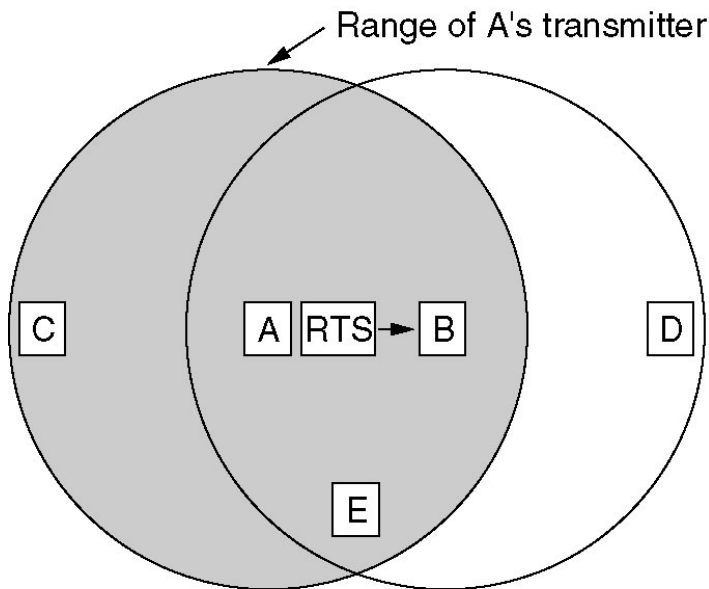
CSMA with Collision Avoidance (MACA)



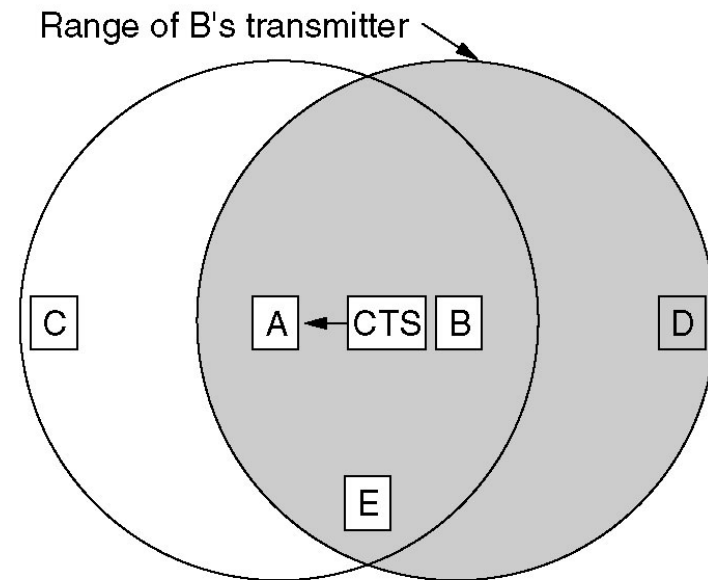
How C knows how long to set NAV (network allocation vector)?

Wireless LAN Protocols - MACA

- **Multiple access with collision avoidance: the sender stimulates the receiver into outputting a short frame so stations nearly can detect the transmission and avoid transmitting for the duration of the upcoming data frame.**



A sending an RTS to B.

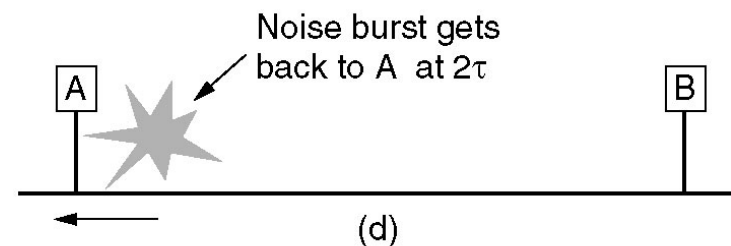
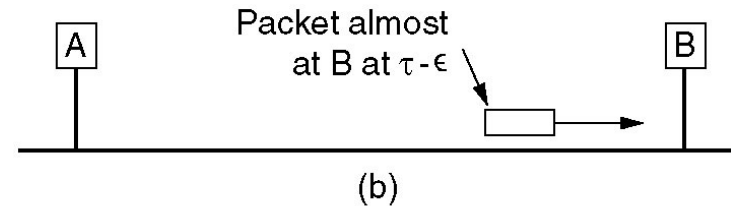
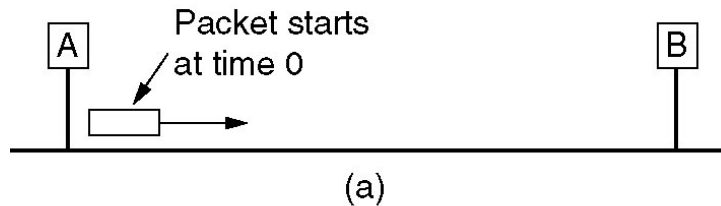


B responding with a CTS to A.

Can C transmit the data somewhere simultaneously? Can D? Can E?
Can a collision still occur?

Ethernet MAC Sublayer Protocol (2)

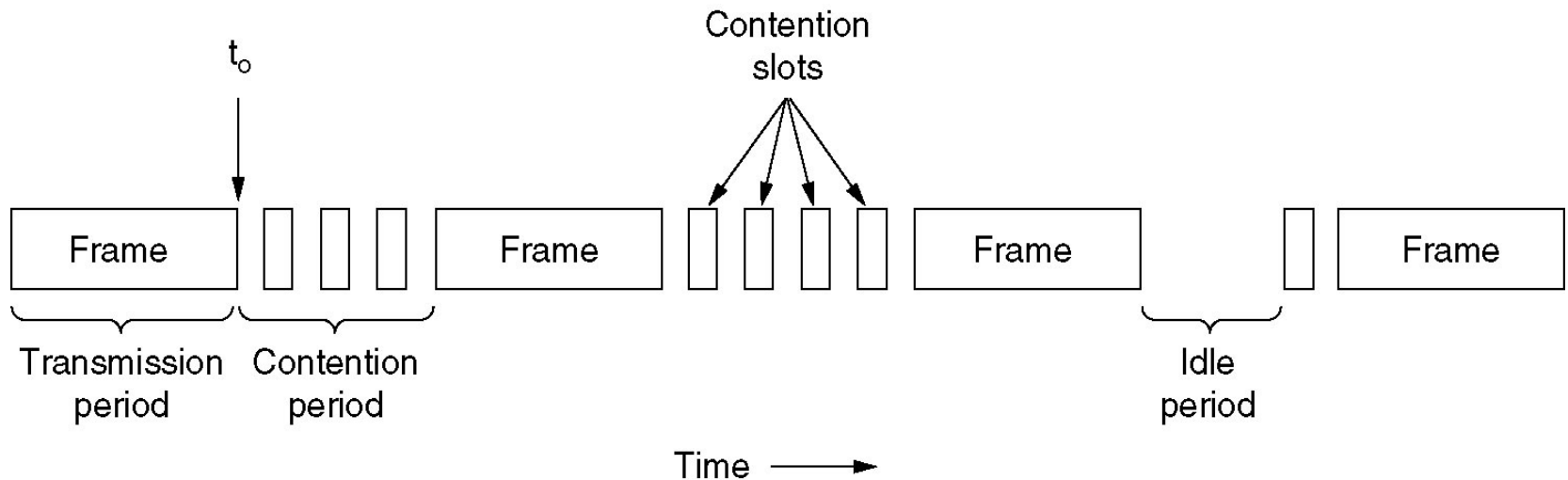
- Why there is a minimum length (64B) for a frame?
 - All frames must take more than 2τ to send so that the transmission is still taking place when the noise burst gets back to the sender.



Collision detection can take as long as 2τ (50 μsec in 10 Mbps 2500m).

The Binary Exponential Backoff Algorithm

- **CSMA/CD: a station aborts its transmission ASAP a detection collision – quickly terminating a damaged frame saves T& BW**
 - If there is a collision, a station waits a random amount of time to try again, how randomization is done?
 - **Binary exponential backoff: after i collisions, a random number between 0 to $2^i - 1$ is chosen, that number of slots is skipped**



CSMA/CD can be in one of three states: contention, transmission, or idle.

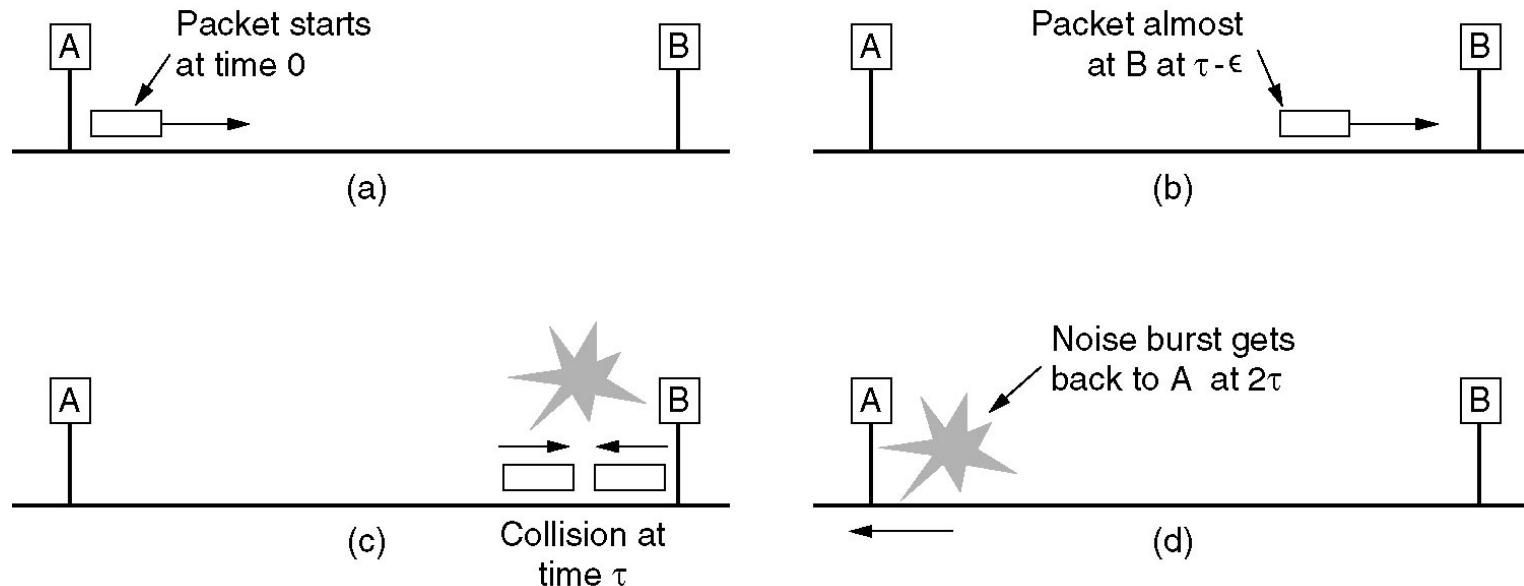
What is a time slot used in Ethernet?

51.2us

Fast Ethernet (2)

- Ethernet frames must be at least 64 bytes long to ensure that the transmitter is still going in the event of a collision at the far end of the cable. Fast Ethernet has the same 64-byte minimum frame size but can get the bits out ten times faster. How is it possible to maintain the same minimum frame size?

The maximum wire length in fast Ethernet is 1/10 as long as in Ethernet.



Collision detection can take as long as 2τ (50 $\rightarrow$ 5 μsec).